



Week 2 :

Regression models and Biased estimators(Regularization)

Kuggle



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1. Regression analysis

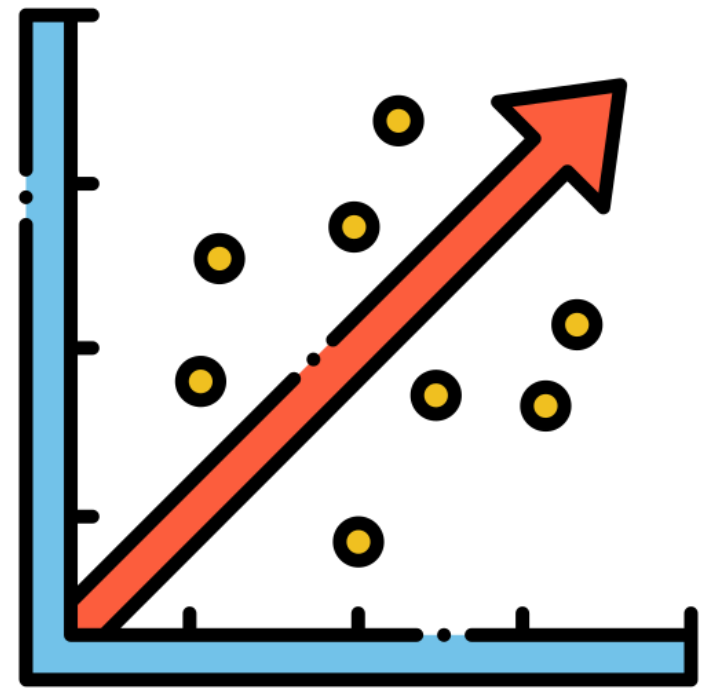
2. Regression evaluation metrics

3. Overfitting and underfitting

4. Regularized linear models – Ridge, Lasso, Elastic Net



1. Regression analysis





1. Regression analysis

Introduction

Explanatory

Result

f

X_1

X_2

$\vdots$

X_n

Y





1. Regression analysis

Introduction

Linear Modeling

$$Y = f(x_1, x_2, x_3, x_4, \dots, x_n)$$

$$Y = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \dots + \beta_p X_p + \varepsilon$$



1. Regression analysis

Introduction

Regression

Model

Simple

Multiple

Linear

$$y = bx + b_0 + \varepsilon$$

Nonlinear

$$y = a \cdot e^{bx} + \varepsilon$$

Linear

$$y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \dots + \beta_p x_p + \varepsilon$$

Nonlinear

$$y = \beta_0 + \beta_1 \cdot x_1 + \beta_2 \cdot x_1^2 + \beta_3 \cdot \sin(x_3) + \varepsilon$$



1. Regression analysis

SLRM

Simple linear regression

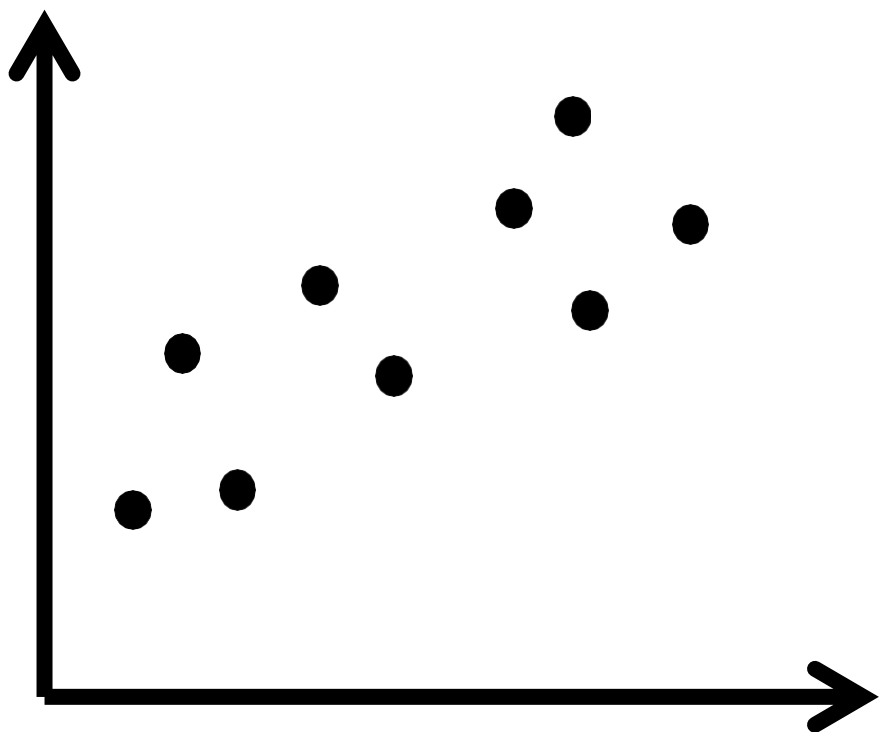
Simple Linear Regression model $Y = \beta_0 + \beta_1 X + \varepsilon$

- β_0 : constant, intercept
- β_1 : slope, coefficient
- ε : error, portion of variation that cannot be explained by the predictors x .



1. Regression analysis

SLRM

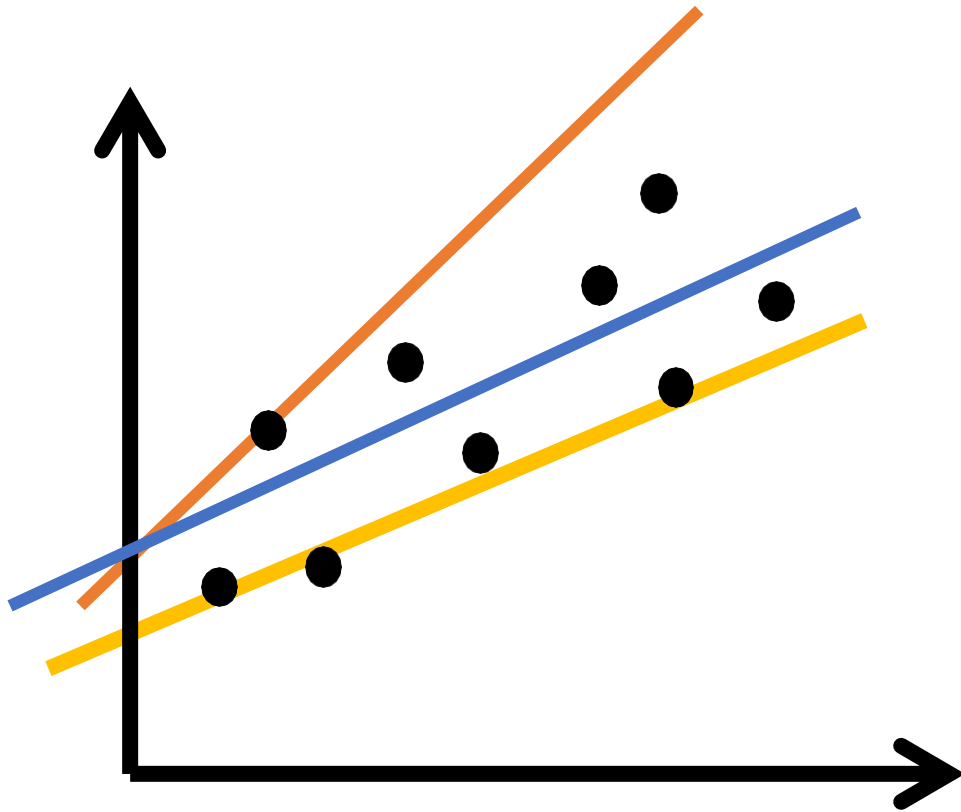




1. Regression analysis

SLRM

Simple linear regression



BlueLine: Regression Line

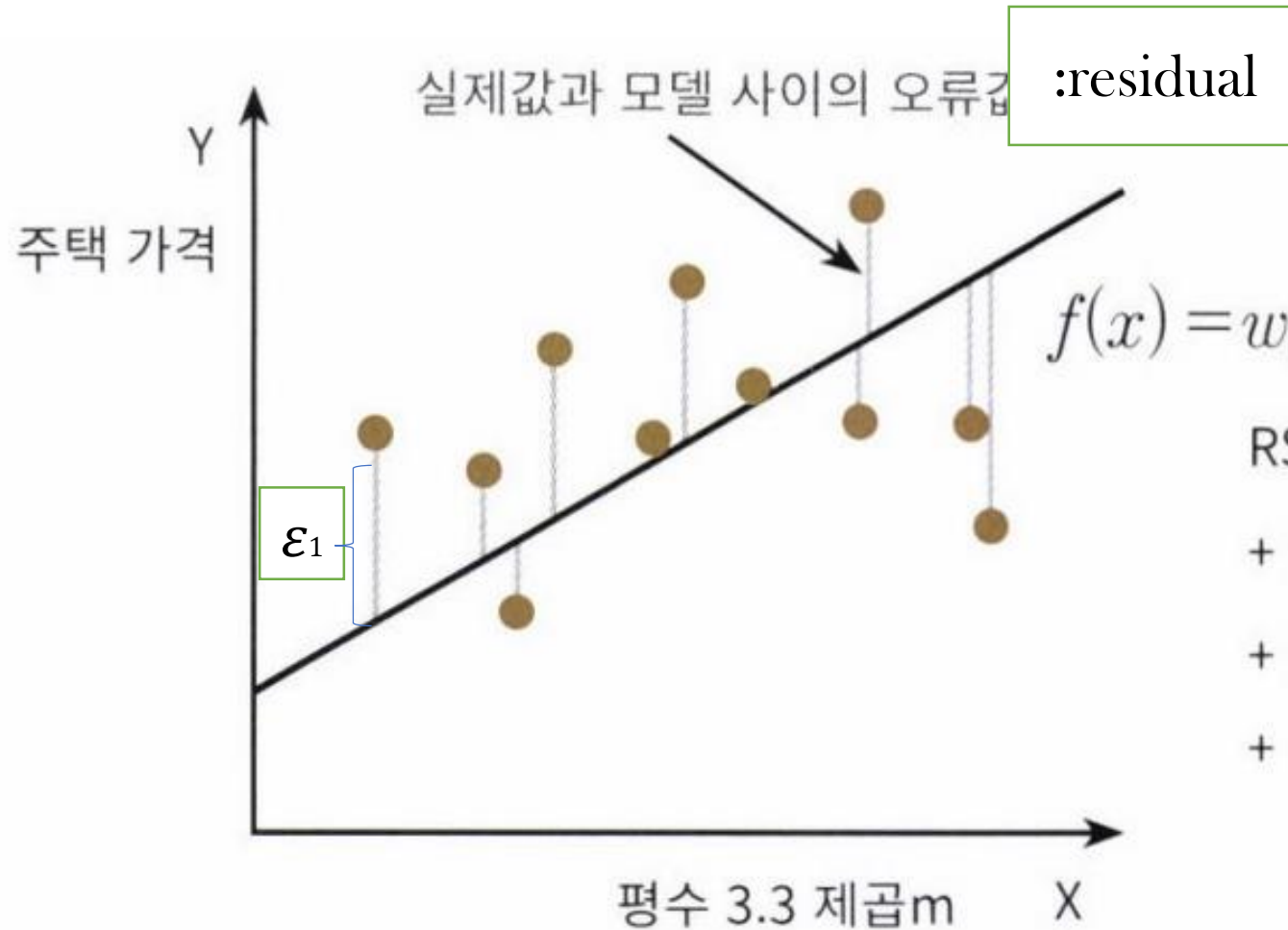
To find the best-fitting line for the data, we measure the distance between the observed values and the candidate regression lines. The line that minimizes this distance is considered the one that best explains the data.

→ This is the core idea behind *least squares*.



1. Regression analysis

RSS



$$\begin{aligned} \text{RSS} = & (\#1 \text{ 주택 가격} - (w_0 + w_1 \text{ \#1 주택크기}))^2 \\ & + (\#2 \text{ 주택 가격} - (w_0 + w_1 \text{ \#2 주택크기}))^2 \\ & + (\#3 \text{ 주택 가격} - (w_0 + w_1 \text{ \#3 주택크기}))^2 \\ & + \dots \text{ (모든 학습 데이터에 대해 RSS 수행)} \end{aligned}$$



1. Regression analysis

RSS

$$RSS(w_0, w_1) = \frac{1}{N} \sum_{i=1}^N (y_i - (w_0 + w_1 * x_i))^2$$

(i 는 1부터 학습 데이터의 총 건수 N 까지)

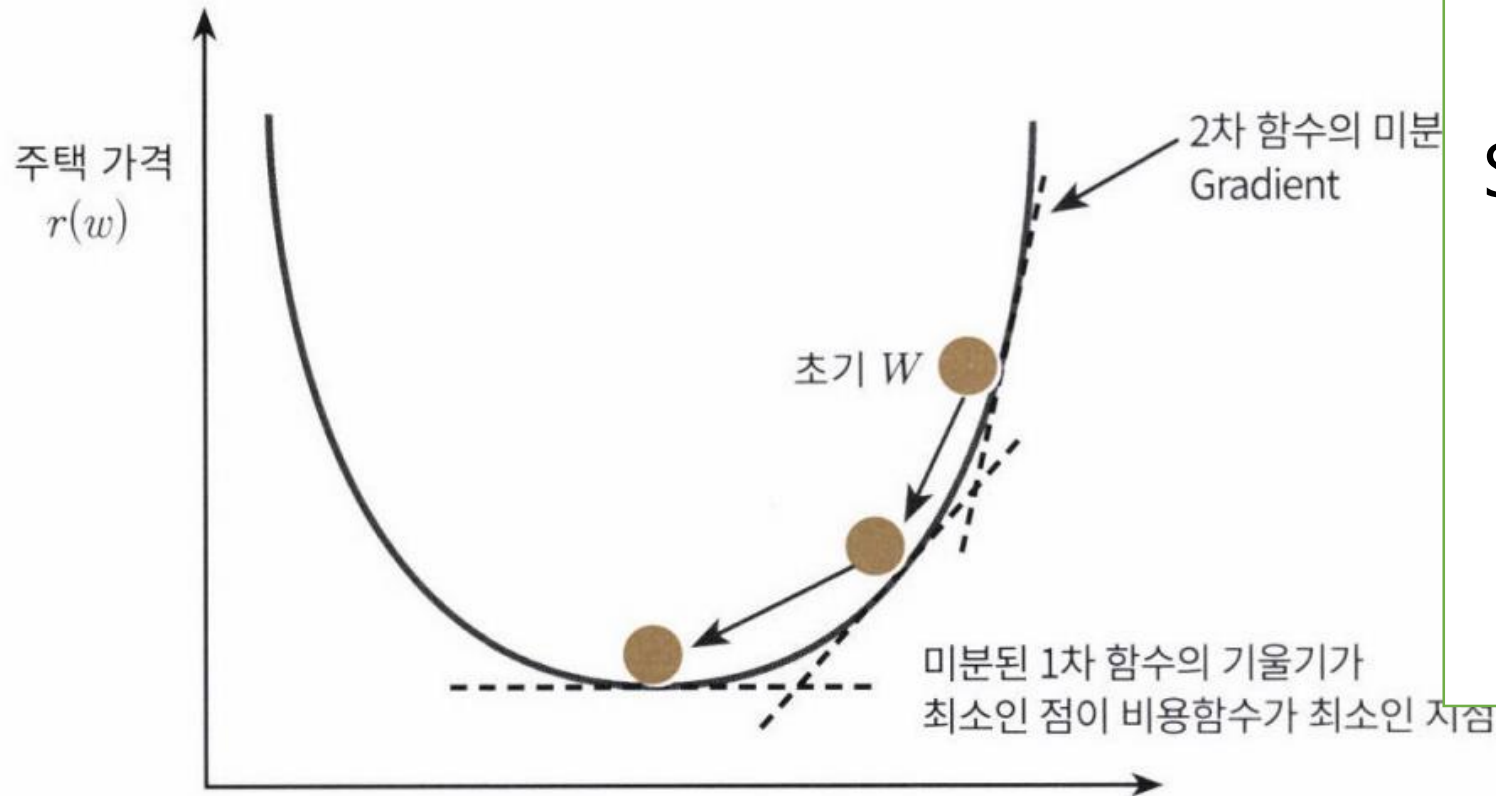
In regression, this measure of discrepancy is called the **Residual Sum of Squares (RSS)**.

The goal of machine-learning-based regression is to learn the parameters w_0 and w_1 that minimize this cost.



1. Regression analysis

Gradient Descent



Sequentially take the partial derivatives of $R(w)$ with respect to w_0 and w_1 , and find the values of w_0 and w_1 that minimize $R(w)$.

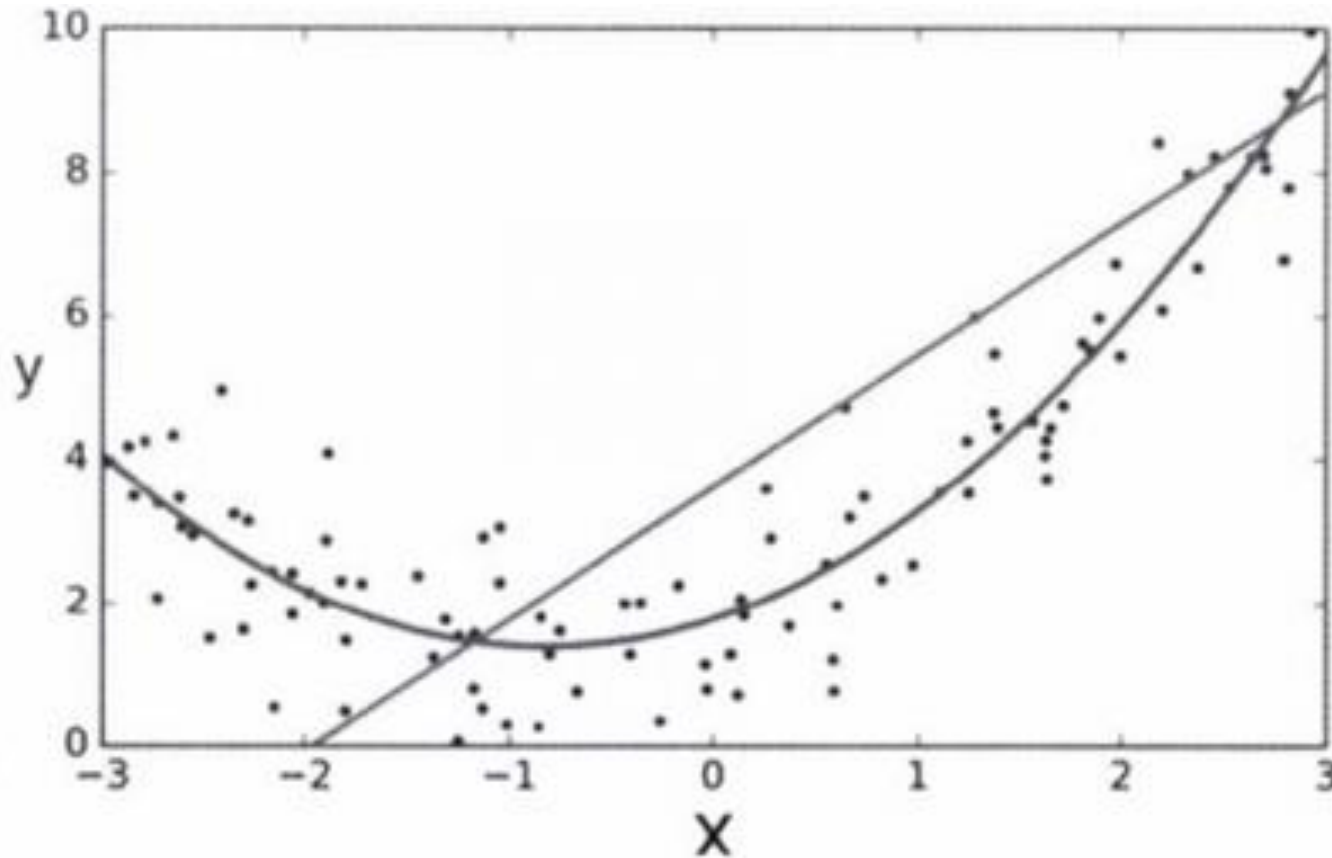
$$\frac{\partial R(w)}{\partial w_1} = \frac{2}{N} \sum_{i=1}^N -x_i * (y_i - (w_0 + w_1 x_i)) = -\frac{2}{N} \sum_{i=1}^N x_i * (\text{실제값}_i - \text{예측값}_i)$$

$$\frac{\partial R(w)}{\partial w_0} = \frac{2}{N} \sum_{i=1}^N -(y_i - (w_0 + w_1 x_i)) = -\frac{2}{N} \sum_{i=1}^N (\text{실제값}_i - \text{예측값}_i)$$



1. Regression analysis

Polynomial Regression



Linear Regression

Polynomial Regression

$$y = w_0 + w_1 * x_1 + w_2 * x_2 + w_3 * x_1 * x_2 + w_4 * x_1^2 + w_5 * x_2^2$$

$$Z = [x_1, x_2, x_1 * x_2, x_1^2, x_2^2]$$

$$y = w_0 + w_1 * z_1 + w_2 * z_2 + w_3 * z_3 + w_4 * z_4 + w_5 * z_5$$



2. Regression evaluation metrics



2. Regression evaluation metrics

Introduction

평가 지표	설명	수식
MAE	Mean Absolute Error(MAE)이며 실제 값과 예측값의 차이를 절대값으로 변환해 평균한 것입니다.	$MAE = \frac{1}{n} \sum_{i=1}^n Y_i - \hat{Y}_i $
MSE	Mean Squared Error(MSE)이며 실제 값과 예측값의 차이를 제곱해 평균한 것입니다.	$MSE = \frac{1}{n} \sum_{i=1}^n (Y_i - \hat{Y}_i)^2$
RMSE	MSE 값은 오류의 제곱을 구하므로 실제 오류 평균보다 더 커지는 특성이 있으므로 MSE에 루트를 씌운 것이 RMSE(Root Mean Squared Error)입니다.	$RMSE = \sqrt{\frac{1}{n} \sum_{i=1}^n (Y_i - \hat{Y}_i)^2}$
R ²	분산 기반으로 예측 성능을 평가합니다. 실제 값의 분산 대비 예측값의 분산 비율을 지표로 하며, 1에 가까울수록 예측 정확도가 높습니다.	$R^2 = \frac{\text{예측값 Variance}}{\text{실제값 Variance}}$



2. Regression evaluation metrics

RMSE

Why do we compute RMSE?

Because the shortcomings of MSE are quite clear. Since MSE squares the errors, the scale (dimension) of the error and the scale of MSE become different. Also, because errors are squared, the value can grow very large.

→ By simply applying a square root, RMSE resolves these issues



2. Regression evaluation metrics

$$R^2$$

Coefficient of Determination R^2

- The R^2 value represents how much of the dependent variable is explained by the regression line.
- $R^2 = \frac{SSR}{SST} = 1 - \frac{SSE}{SST}$
- $0 \leq R^2 \leq 1$

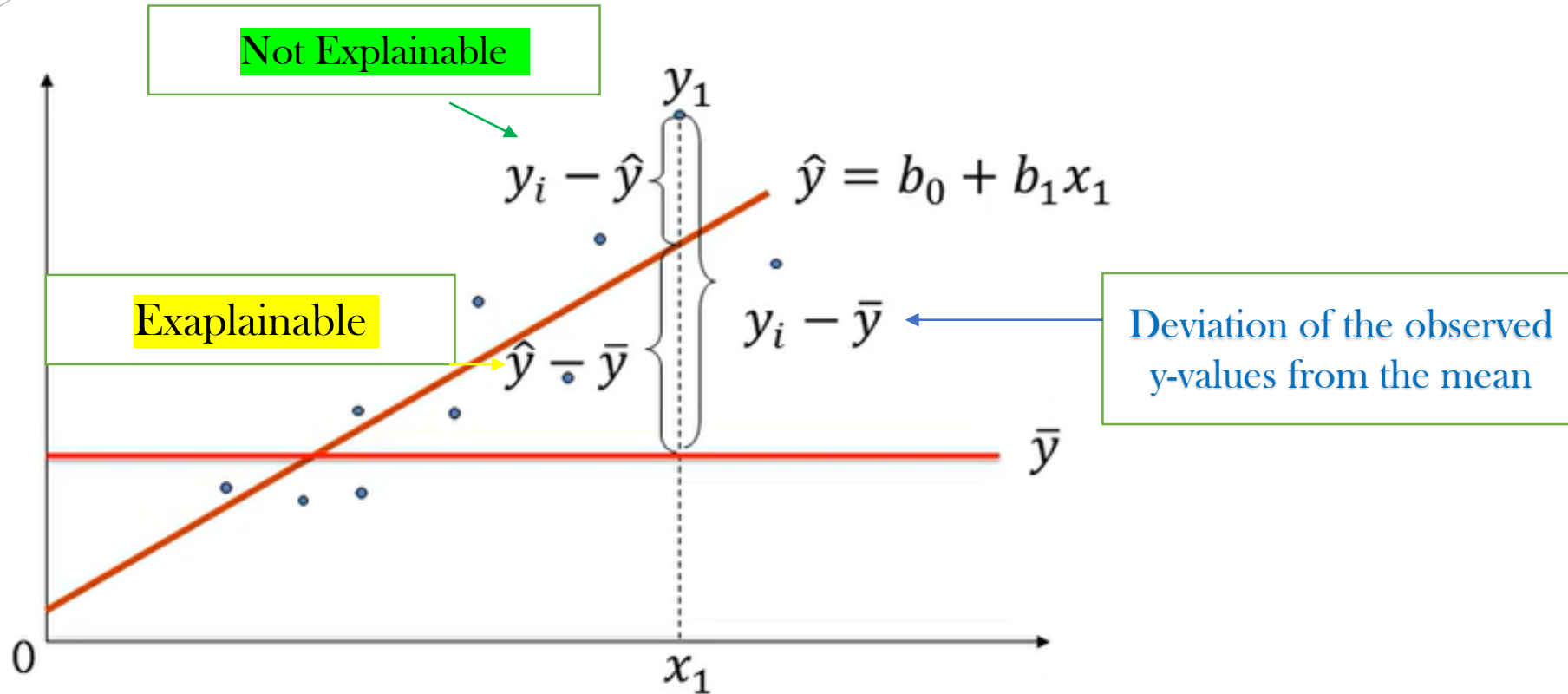
(Close to **0** → the regression line does *not* explain the data well,

Close to **1** → the regression line explains the data *well*)



2. Regression evaluation metrics

Decomposition of SSE



$$\sum (y_i - \bar{y})^2 = \sum (\hat{y} - \bar{y})^2 + \sum (y_i - \hat{y})^2$$

$$SST = SSR + SSE$$



2. Regression evaluation metrics

Codes & Application

```
import numpy as np
import matplotlib.pyplot as plt
import pandas as pd
import seaborn as sns
from scipy import stats
from sklearn.datasets import fetch_california_housing
from sklearn.datasets import fetch_openml
%matplotlib inline

# california 데이터셋 로드
housing = fetch_openml(name="house_prices", as_frame=True)

# california 데이터셋 DataFrame 변환
california = fetch_california_housing(as_frame=True)
californiaDF = california['frame']
californiaDF.head()
```

Visualization

Stats function

Dataset of California Housing



2. Regression evaluation metrics

Codes & Application

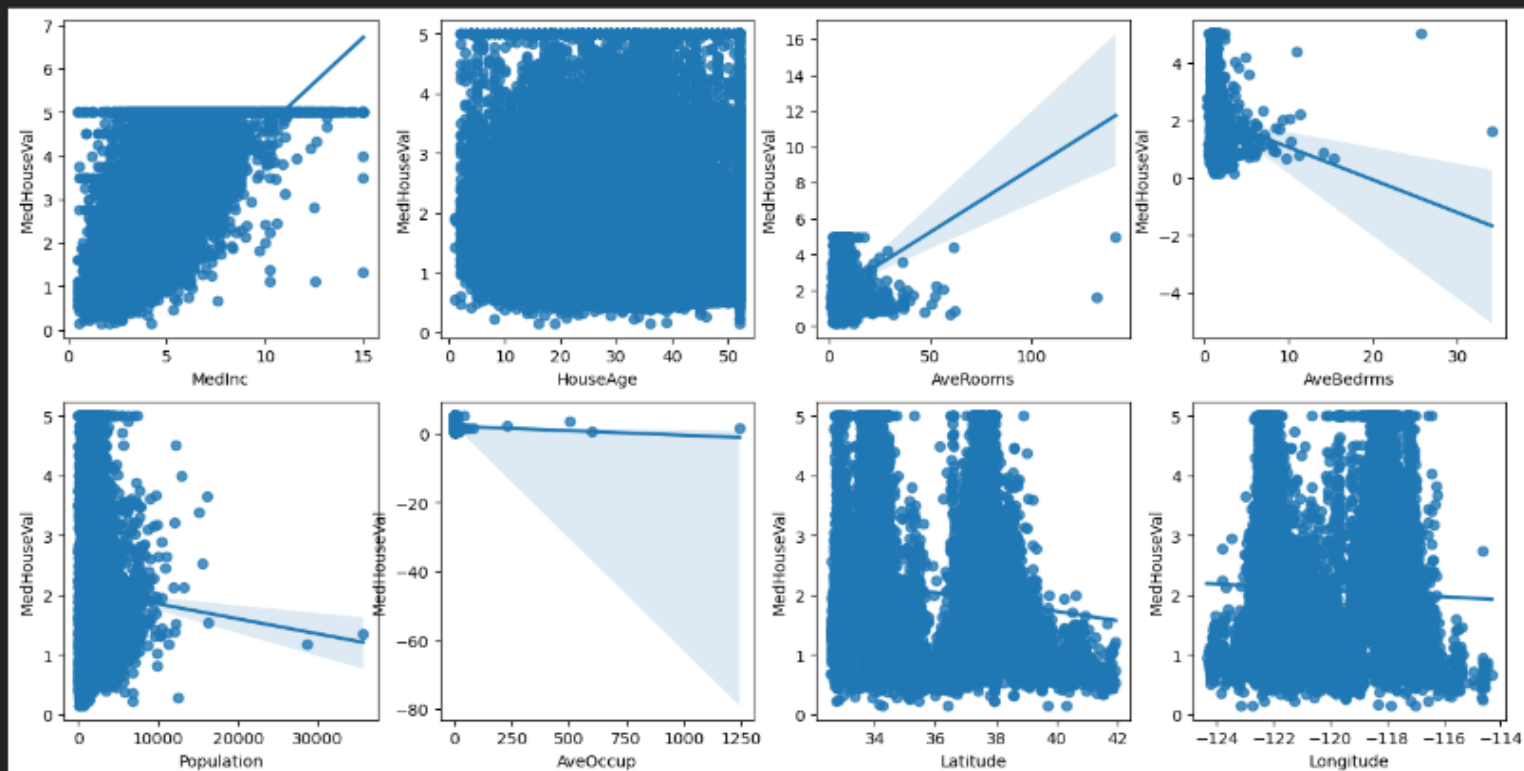
Variables	
MedInc	Median income
House Age	Median house age
Ave Rooms	Average number of rooms
Ave Bedrms	Average number of bedrooms
Population	
Ave Occup	Average household occupancy
Latitude	
Longitude	

	MedInc	HouseAge	AveRooms	AveBedrms	Population	AveOccup	Latitude	Longitude	MedHouseVal
0	8.3252	41.0	6.984127	1.023810	322.0	2.555556	37.88	-122.23	4.526
1	8.3014	21.0	6.238137	0.971880	2401.0	2.109842	37.86	-122.22	3.585
2	7.2574	52.0	8.288136	1.073446	496.0	2.802260	37.85	-122.24	3.521
3	5.6431	52.0	5.817352	1.073059	558.0	2.547945	37.85	-122.25	3.413
4	3.8462	52.0	6.281853	1.081081	565.0	2.181467	37.85	-122.25	3.422

2. Application

Codes & Application

```
# 2개의 행과 4개의 열을 가진 subplots를 이용. axs는 4x2개의 ax를 가짐.  
fig, axs = plt.subplots(figsize=(16, 8), ncols=4, nrows=2)  
  
lm_features = ['MedInc', 'HouseAge', 'AveRooms', 'AveBedrms', 'Population', 'AveOccup',  
              'Latitude', 'Longitude']  
  
for i, feature in enumerate(lm_features):  
    row = int(i / 4)  
    col = i % 4  
    # 시본의 regplot을 이용해 산점도와 선형 회귀 직선을 함께 표현  
    sns.regplot(x=feature, y='MedHouseVal', data=californiaDF, ax=axs[row][col])
```





3. Overfitting and underfitting

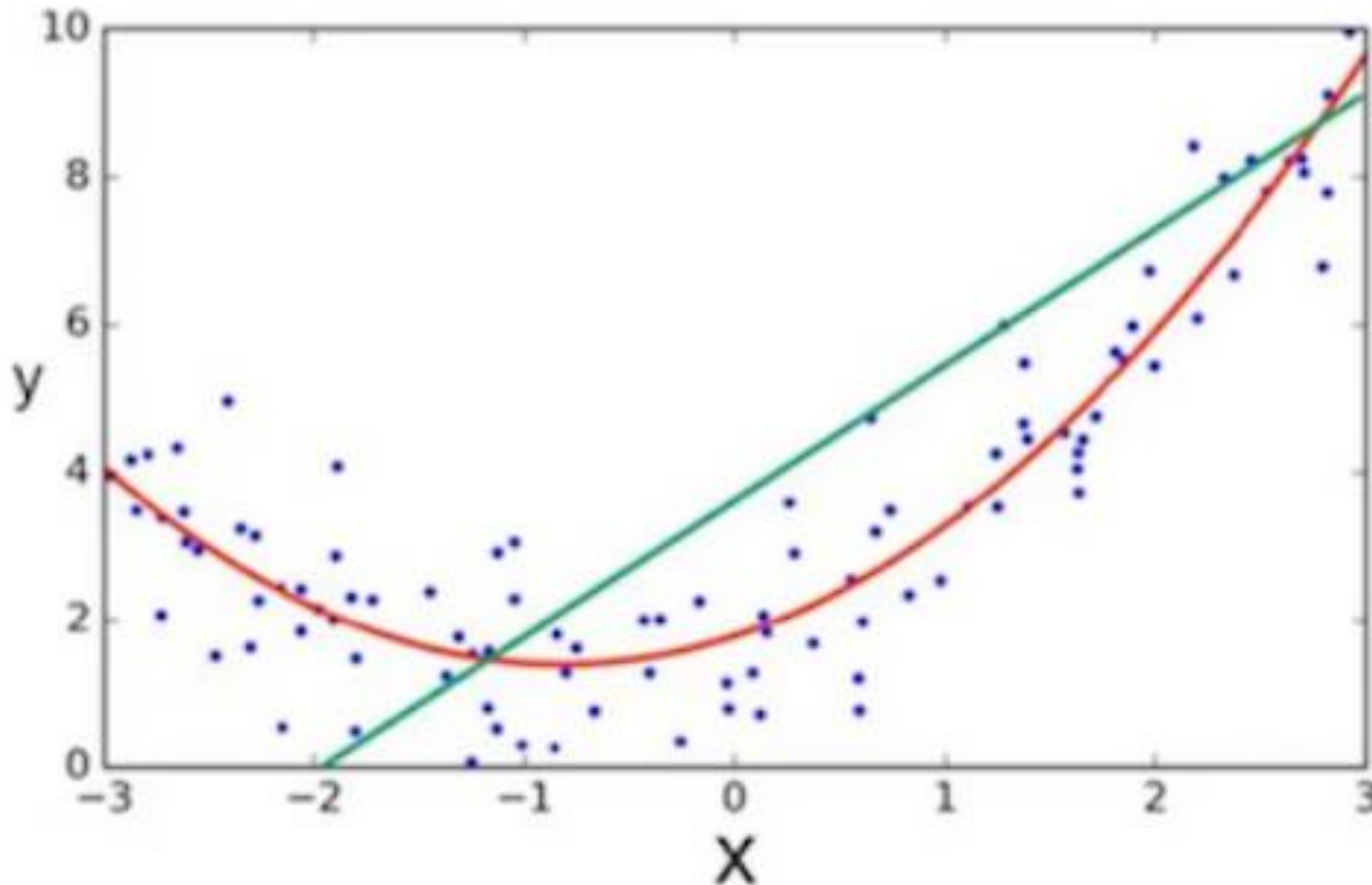




3. Overfitting and underfitting

Polynomial Regression

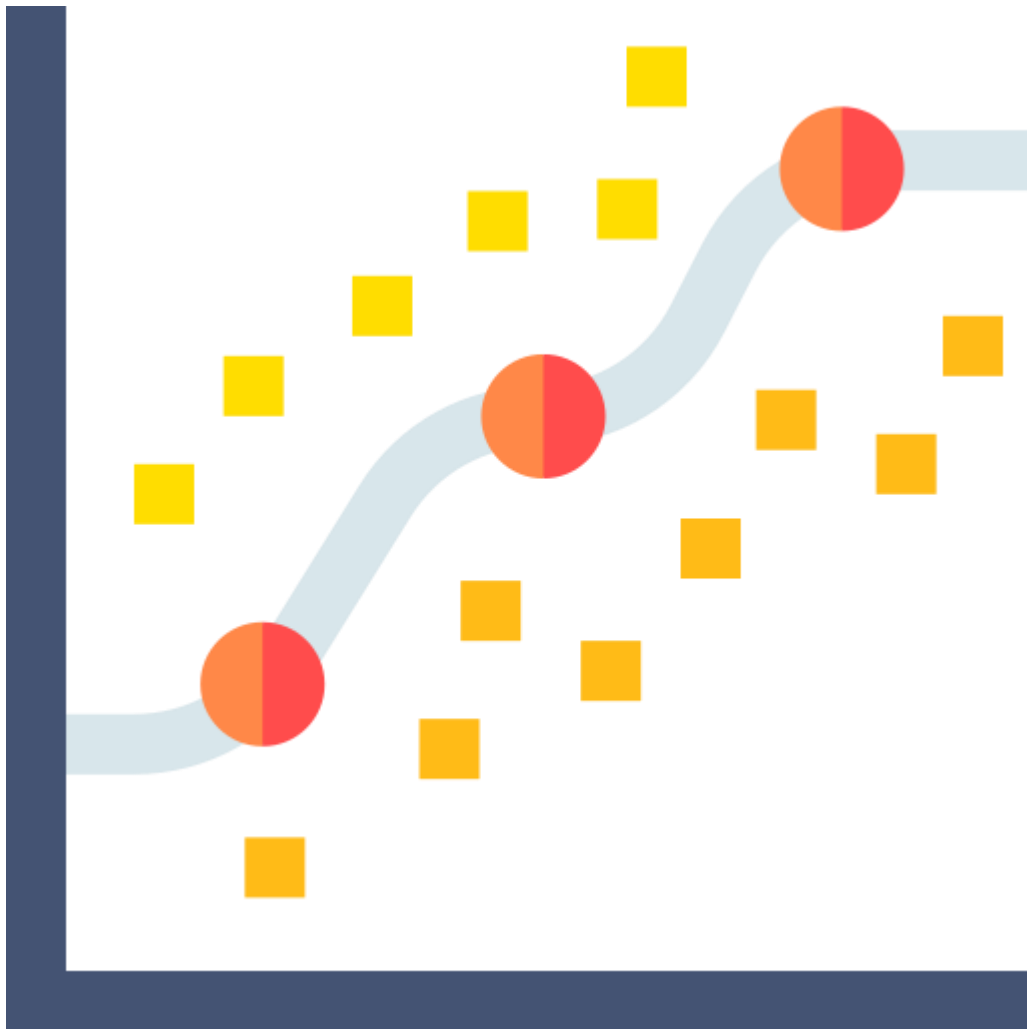
Is the Appropriate Model a Line or a Curve?





3. 과(대)/과소 적합

다항회귀



Polynomial Regression

The independent variable is expressed as a polynomial. Models complex nonlinear relationships.

Choosing an appropriate degree is crucial.

Advantages:

Can model complex features and relationships.

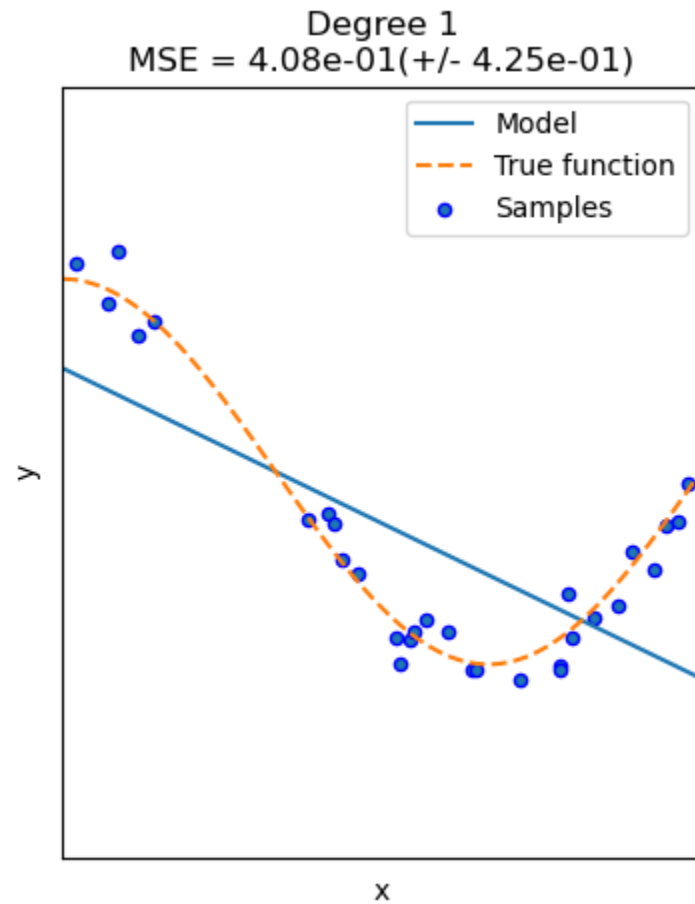
Disadvantages:

Prediction accuracy may decrease if overly complex.

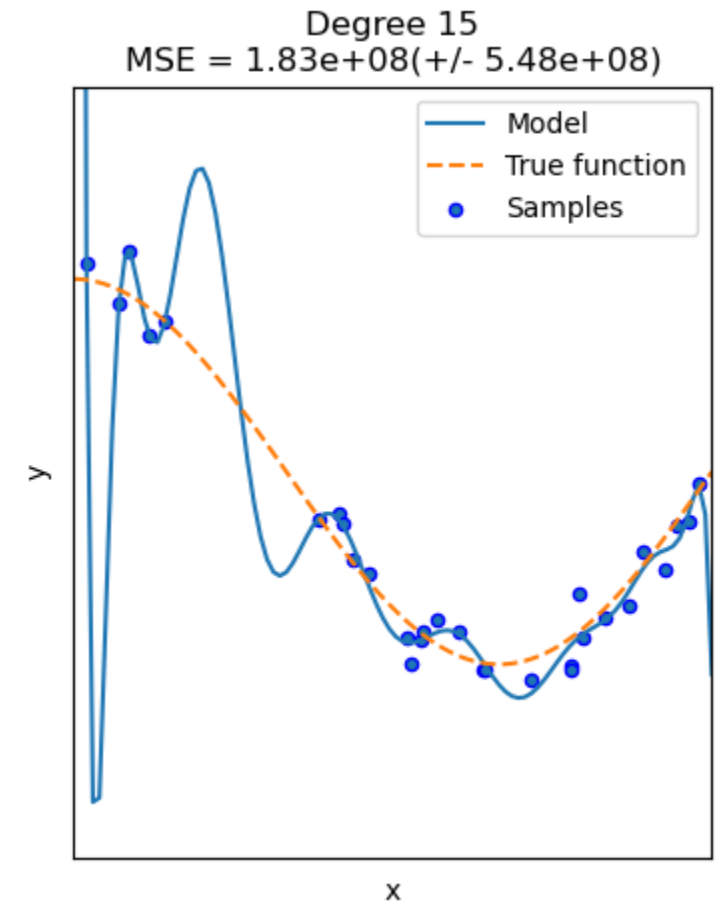
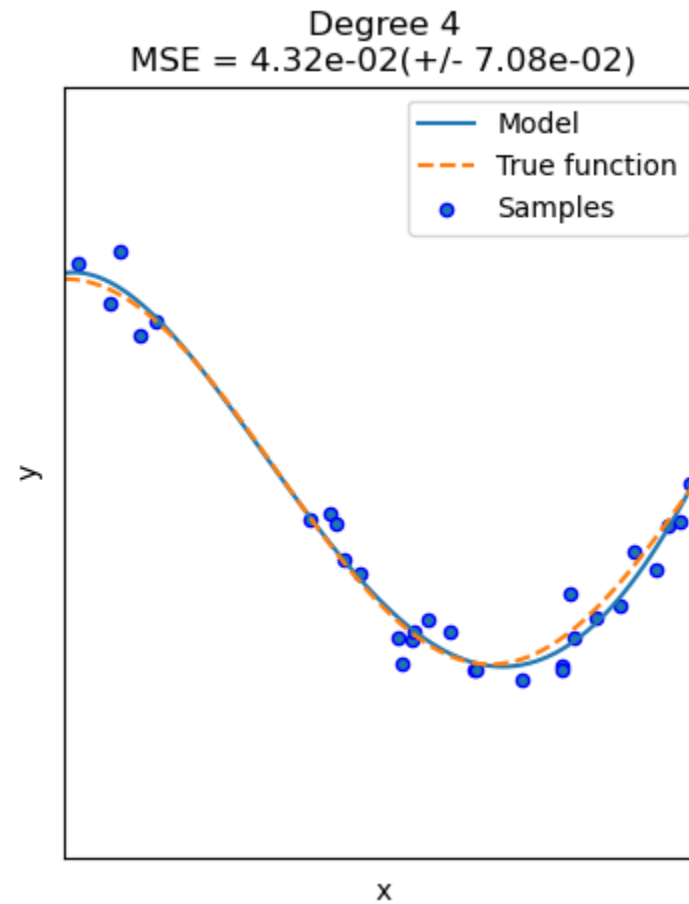


3. Overfitting and underfitting

Polynomial Degree



High Bias



High Variance



3. Overfitting and underfitting

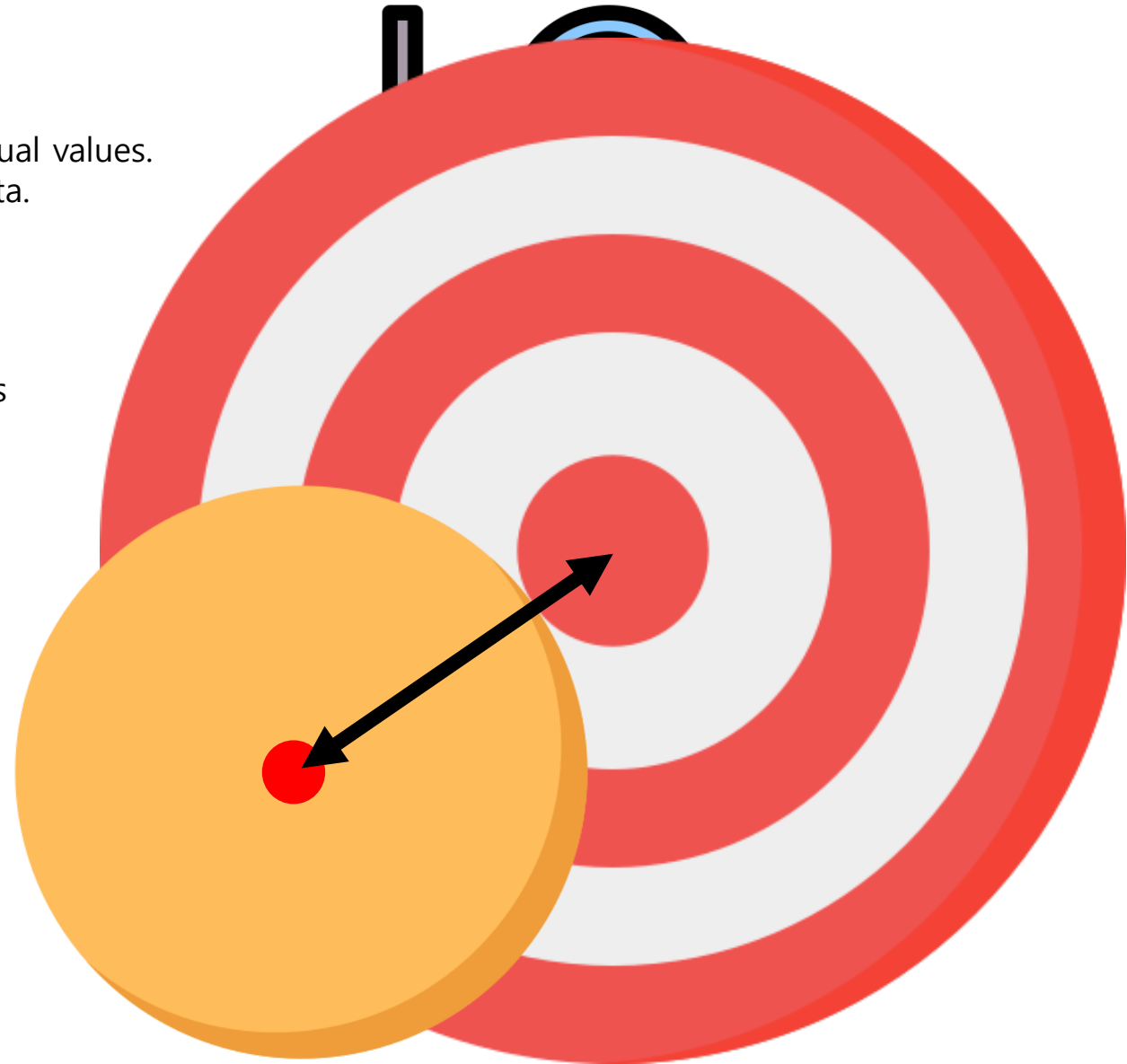
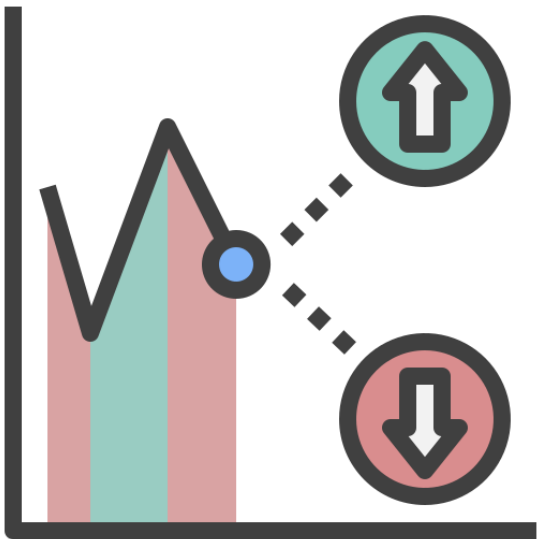
Bias, Var

Bias

Bias is the difference between the model's predicted values and the actual values. It represents how far the average of the predictions is from the true data.

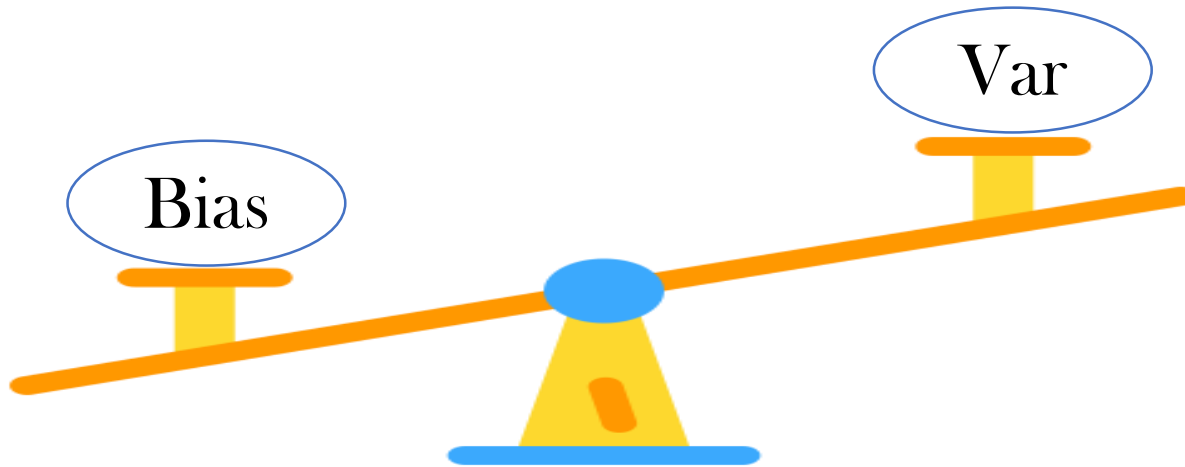
Variance

Variance refers to the variability of predictions when the same model is trained on different datasets. It measures how sensitive the model is to fluctuations in the data.

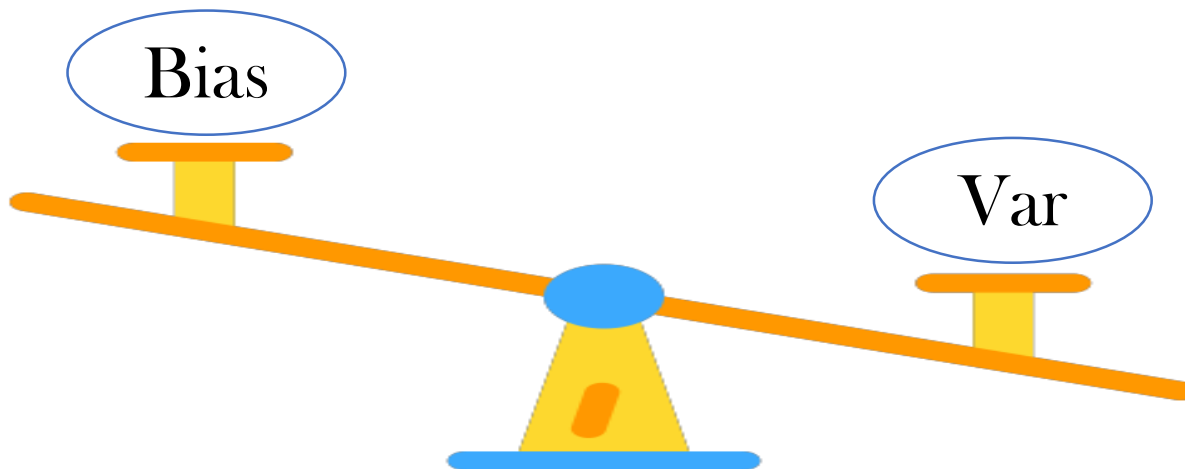




3. Overfitting and underfitting



Overfitting



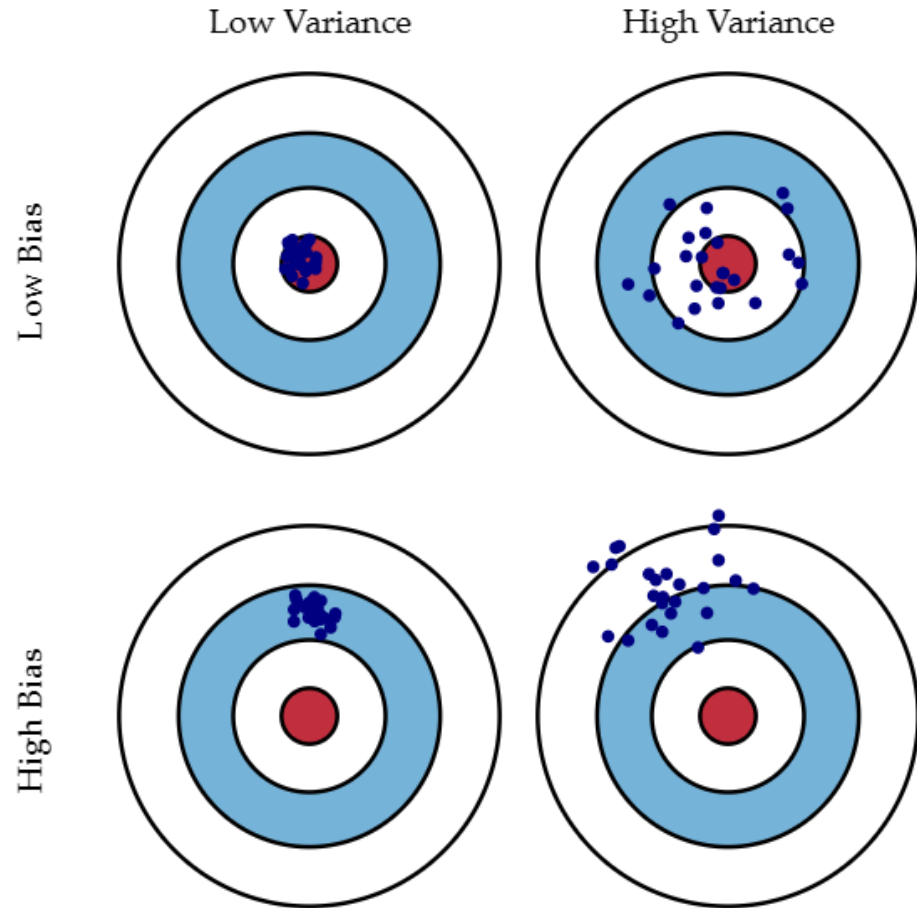
Underfitting



3. Overfitting and underfitting

dilemma

Var-Bias Trade off



Degree 1 Model

- Very simplified model
- Strong directional tendency
- **High bias**

Degree 15 Model

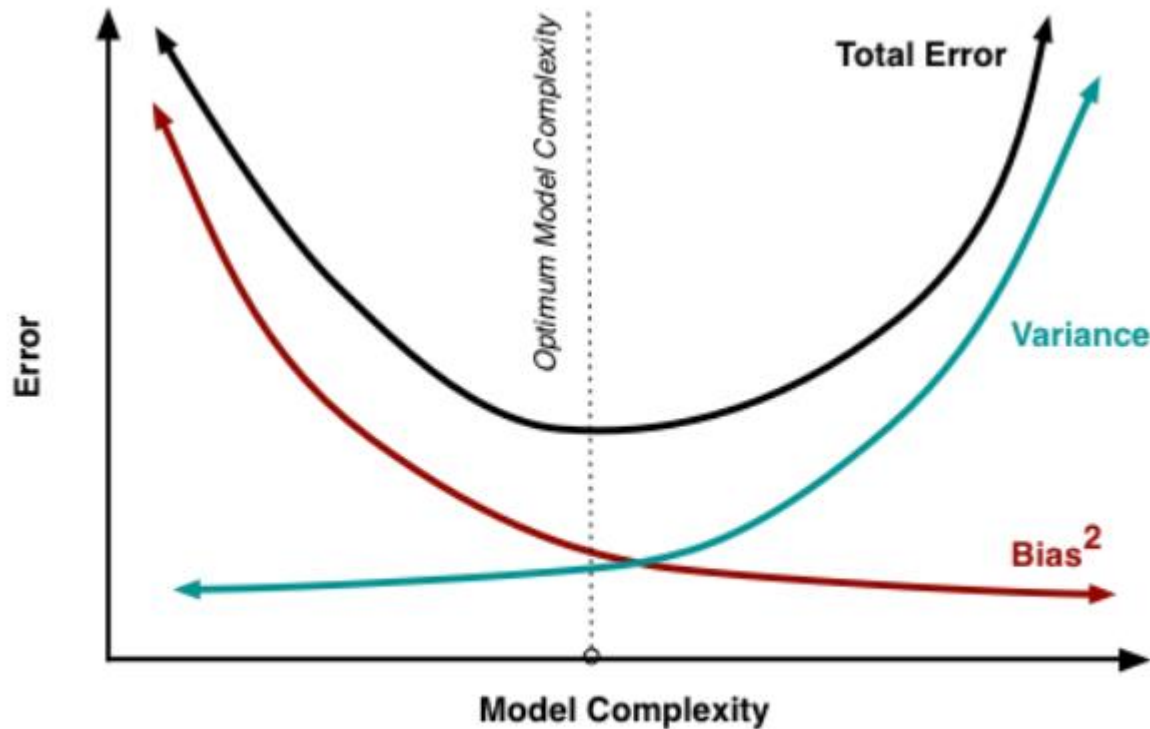
- Extremely complex model
- Very high variability
- **High variance**

Fig. 1 Graphical illustration of bias and variance.



3. Overfitting and underfitting

Goldilocks Point



The “Goldilocks” Point

- The point of optimal balance
- Bias is reduced and variance is controlled
- Overall error becomes the lowest

Fig. 6 Bias and variance contributing to total error.

<https://scott.fortmann-roe.com/docs/BiasVariance.html>

Kuggle

4. Regularized linear models - Ridge, Lasso, Elastic Net





4. Regularized linear models

- Ridge, Lasso, Elastic Net



Reducing error for
Training data

Coefficient
complexity



Optimizing
cost function

Balance!



4. Regularized linear models

- Ridge, Lasso, Elastic Net

(Curse of Dimensionality)



As Dim gets larger
(Feature increases)



RSS reduces



Impact on regression coeff



Overfitting, low performance on predictive error

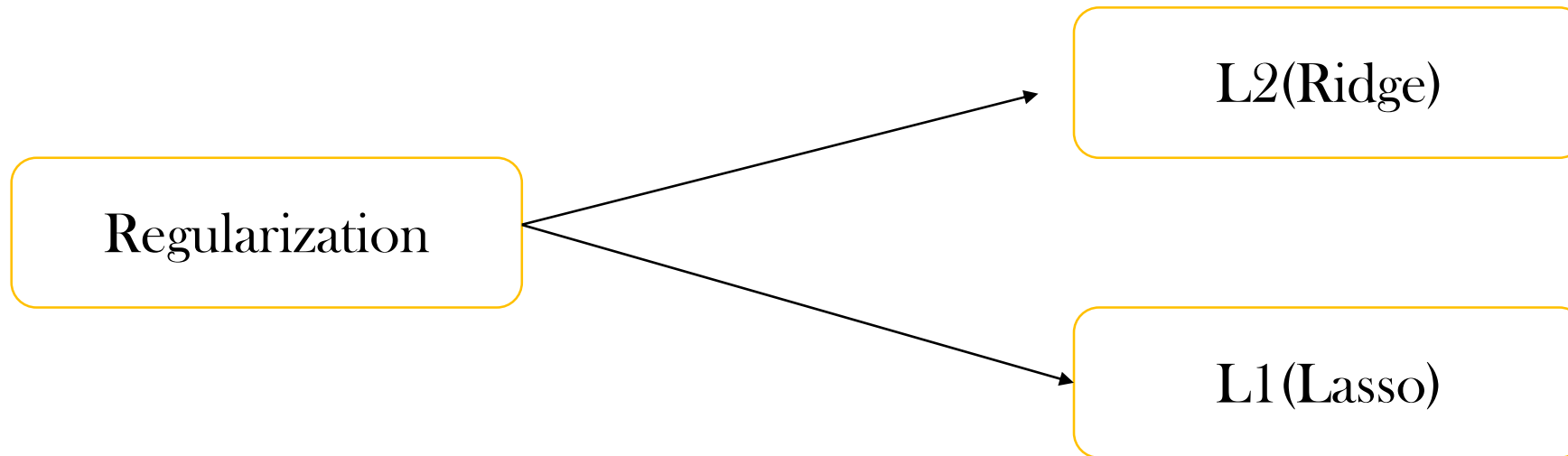


4. Regularized linear models

- Ridge, Lasso, Elastic Net

$$\text{비용 함수 목표} = \text{Min}(\text{RSS}(W) + \alpha * \|W\|_2^2)$$

Regularization reduces overfitting by adding a penalty term (controlled by **alpha**) to the cost function.





4. Regularized linear models

- Ridge, Lasso, Elastic Net

Ridge regression

```
# 필요한 라이브러리 임포트
from sklearn.model_selection import train_test_split
from sklearn.linear_model import Ridge
from sklearn.metrics import mean_squared_error

# 데이터 로드
df = load_df()
X, y = df.data, df.target

# 데이터 분할 (훈련 세트와 테스트 세트로)
X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.3, random_state=42)

# 릿지 회귀 모델 생성 및 학습
ridge = Ridge(alpha=1.0) # alpha는 규제 강도를 조절하는 파라미터
ridge.fit(X_train, y_train)

# 예측 수행
y_pred = ridge.predict(X_test)

# 성능 평가 (RMSE 계산)
mse = mean_squared_error(y_test, y_pred)
rmse = mse ** 0.5 # or np.sqrt(mse)
print('RMSE: ', rmse)
```



4. Regularized linear models

- Ridge, Lasso, Elastic Net

Lasso regression

```
# 필요한 라이브러리 импорт
from sklearn.model_selection import train_test_split
from sklearn.linear_model import Lasso
from sklearn.metrics import mean_squared_error

# 데이터 로드
df = load_df()
X, y = df.data, df.target

# 데이터 분할 (훈련 세트와 테스트 세트로)
X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.3, random_state=42)

# 라쏘 회귀 모델 생성 및 학습
lasso = Lasso(alpha=1.0) # alpha는 규제 강도를 조절하는 파라미터
lasso.fit(X_train, y_train)

# 예측 수행
y_pred = lasso.predict(X_test)

# 성능 평가 (RMSE 계산)
mse = mean_squared_error(y_test, y_pred)
rmse = mse ** 0.5 # or np.sqrt(mse)
print('RMSE: ', rmse)
```



4. Regularized linear models

- Ridge, Lasso, Elastic Net

Elastic net regression

Lasso Regression

When predictors are highly correlated, Lasso tends to select only the important features. It often shrinks the coefficients of other variables to zero.



Coefficient values change sharply depending on alpha.

$$\text{RSS}(W) + \alpha_2 * \|W\|_2^2 + \alpha_1 * \|W\|_1$$

* 엘라스틱 넷에서 규제

$$a \cdot L1 + b \cdot L2$$

* 엘라스틱 넷에서 α 파라미터

$$a+b$$

* 엘라스틱 넷에서 $L1$ -ratio 파라미터

$$\frac{a}{a+b}$$

if) i) $L1\text{-ratio} = 0 \rightarrow L2$ 규제
ii) $L1\text{-ratio} = 1 \rightarrow L1$ 규제



4. Regularized linear models

- Ridge, Lasso, Elastic Net

Elastic net regression

```
# 필요한 라이브러리 임포트
from sklearn.model_selection import train_test_split
from sklearn.linear_model import ElasticNet
from sklearn.metrics import mean_squared_error

# 데이터 로드
df = load_df()
X, y = df.data, df.target

# 데이터 분할 (훈련 세트와 테스트 세트로)
X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.3, random_state=42)

# 엘라스틱넷 회귀 모델 생성 및 학습
# alpha는 규제 강도를 조절하는 파라미터이고 l1_ratio는 L1 규제의 비율을 의미
elastic_net = ElasticNet(alpha=1.0, l1_ratio=0.5)
elastic_net.fit(X_train, y_train)

# 예측 수행
y_pred = elastic_net.predict(X_test)

# 성능 평가 (RMSE 계산)
mse = mean_squared_error(y_test, y_pred)
rmse = mse ** 0.5 # or np.sqrt(mse)
print('RMSE: ', rmse)
```



2주차 과제

과제

1. $y=10+9X+e$ 에 해당하는 자료(산점도) 구성하기.

2. y에는 'MedHouseVal', X에는 'MedHouseVal'를 제외한 나머지 피처를 이용하여 다중회귀 모델 만들기

3. 릿지회귀

```
15):  
# 앞의 Lin  
from sklearn  
from sklearn  
  
#릿지회귀  
ridge =  
#교차검증  
neg_mse_sc
```

4. MAE가 아닌 RMSE를 사용하는 이유는 무엇인가요?

MAE는 실제 값과 예측값의 차이를 절댓값으로 변환해 평균낸 것입니다. MAE는 다른 지표들에 비해 직관적이라는 특징을 가지고 있는데, 이러한 장점을 가진 MAE가 아닌 RMSE를 사용하는 이유는 무엇인지 설명해주세요.

● 참고자료: <https://data101.oopy.io/mae-vs-rmse>

5.편향-분산 트레이드 오프에 대해서 자세히 설명후 해결방안을 찾아주세요